RANK-ONE HORROCKS OVER ANY COMMUTATIVE RING, BY A LOCAL-GLOBAL PRINCIPLE FOR COMAXIMAL MONOIDS

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Abstract. We give a constructive proof that an ideal of $A[X]$ dividing a principal ideal generated by a monic polynomial is principal, for an arbitrary commutative ring A. The local part follows Nashier–Nichols. The global part uses the comaximal monoids produced by a formal gcd computation directly, through two general lemmas: membership in a finitely generated ideal is local, and compatible local elements glue. The gluing lemma has an explicit proof without principal localisations and without exponents. This organisation of the proof was suggested by writing a complete machine-checked version.

1. Statement

Let A be a commutative ring. All rings are commutative.

Theorem 1. *Let $F \in A[X]$ be monic, and $y_1, \dots, y_n$, $x_1, \dots, x_n \in A[X]$ such that*

$$\textstyle\sum_i x_i y_i = F, \qquad x_i y_j = F e_{ij} \quad (1 \le i, j \le n)$$

for some $e_{ij} \in A[X]$. Then the ideal $L = (y_1, \dots, y_n)$ is principal. Moreover there are $a_i, b_i \in A[X]$ with $b_i a_j = e_{ij}$ and $\sum_i a_i b_i = 1$.

The hypothesis says $L \cdot M = (F)$ for $M = (x_1, \dots, x_n)$. If e is an idempotent matrix of rank one over $A[X]$ which becomes free over $A[X]_f$, with f monic, then there are x, y satisfying the hypothesis with $F = f^k$, and the conclusion says that e is free. This is the rank-one case of Horrocks' theorem used in [1]; the projector equations themselves are not needed. The generator of L need not be monic: if $\varepsilon^2 = \varepsilon$ then $(\varepsilon X + 1 - \varepsilon)$ divides (X).

We write $S^{-1}A$ or A_S for the localisation at a multiplicative monoid S. Recall that $a/1 = 0$ in A_S means $sa = 0$ for some $s \in S$.

2. Comaximal monoids

For finite sets $I, U \subseteq A$, let $M(U)$ be the monoid generated by U and

$$S(I;U) = M(U) + (I).$$

This is a monoid, and $S(I;U) + (I) = S(I;U)$, so the ideal generated by I in $A_{S(I;U)}$ is contained in its Jacobson radical: for $c \in (I)$ and $s \in S$, $1 + c/s = (s + c)/s$ is invertible.

A *tree* is a finite binary tree whose inner nodes are labelled by elements of A. Its nodes carry pairs $(I;U)$: the root carries $(\emptyset;\emptyset)$ and a node $(I;U)$ labelled a has the children $(I, a; U)$ and $(I; U, a)$.

Lemma 2 ([3, Chapter XV])**.** *Choose, for each leaf $(I;U)$ of a tree, an element s of $S(I;U)$. Then the chosen elements generate the unit ideal.*

Proof. Let $\mathfrak{a}$ be the ideal generated by the chosen elements. We show by induction on the tree that $\mathfrak{a}$ meets $S(I;U)$ for every node; at the root $S(\emptyset;\emptyset) = \{1\}$. It suffices to show: if $s \in S(I, a; U)$ and $t \in S(I; U, a)$, then (s, t) meets $S(I;U)$. Write $s = u + i + ab$ and $t = u' a^k + i'$

with $u, u' \in M(U)$ and $i, i' \in (I)$. Expanding $u^k = (s - i - ab)^k$ gives $u^k = cs + j + (-b)^k a^k$ with $j \in (I)$, hence

$$cu's + (-b)^k t = u'u^k + (-u'j + (-b)^k i') \in S(I;U). \qquad \square$$

The proof never decides anything about elements of A. A finite family of monoids with this property is called *comaximal*.

3. Local generators

In this section R is a ring, J an ideal contained in its Jacobson radical, and we write $\bar{}$ for reduction modulo J. A monic polynomial is regular, also modulo J.

Lemma 3 (Monic lifting [2, IV.5.1], [1, Lemma 7.5])**.** *Let $L \subseteq R[X]$ be an ideal containing a monic polynomial. If $\rho \in L$ has all coefficients of degree $> r$ in J and its coefficient of degree r invertible modulo J, then L contains a monic polynomial h of degree r with $\bar{h} = \bar{\rho}\,\bar{c}$ for some unit c.*

Lemma 4 ([2, IV.5.2], [1, Lemma 7.6])**.** *Let y, x, e, F over $R[X]$ satisfy the hypothesis of Theorem 1, and let $h = \sum_j c_j y_j$ be monic with $y_i - ha_i \in J[X]$ for all i. Then $y_i \in (h)$ for all i, so $L = (h)$.*

Proof. First, if $p = \sum_j c'_j y_j$ lies in $J[X]$ then $p \in J[X]\,L$. Indeed put $q_i = \sum_j e_{ij} c'_j$, so that $x_i p = F q_i$. Then $\bar{F}\bar{q}_i = 0$, hence $q_i \in J[X]$, and $Fp = \sum_i y_i x_i p = F \sum_i q_i y_i$ gives $p = \sum_i q_i y_i$. Applied to $p = y_i - ha_i$ this gives

$$y_i = ha_i + \textstyle\sum_j q_{ij} y_j, \qquad q_{ij} \in J[X].$$

Let $d = \deg h$ and let v_i be the image of y_i in $R[X]/(h)$. The R-module N generated by the $X^k v_i$, $k < d$, is closed under multiplication by X, since h is monic. From $v_i = \sum_j q_{ij} v_j$ we get $N = JN$, and N is finitely generated, so $N = 0$ by Nakayama's lemma. Hence $y_i \in (h)$. $\square$

Lemma 5. *If $h, h' \in R[X]$ are monic of degrees d, d' and $(h) = (h')$, then $h = h'$ if $d = d'$, and $R = 0$ if $d \neq d'$.*

Proof. A nonzero multiple of a monic polynomial of degree d has degree at least d: comparing coefficients from the top shows that a multiple of degree $< d$ is 0. Write $h = h'a$. If $d < d'$ (the case $d > d'$ is symmetric), then $h = 0$ by this remark, and $1 = 0$ since h is monic. If $d = d'$, write also $h' = hb$; then $h - h' = h'(a - 1)$ has degree $< d'$, so $h = h'$. $\square$

4. Formal gcd

We keep the hypothesis of Theorem 1. For a node $(I;U)$ write $S = S(I;U)$ and $J = (I)A_S$.

Lemma 6. *There is a tree such that at every leaf $(I;U)$ there is a monic $h \in LA_S[X]$ with $y_i - ha_i \in J[X]$ for some $a_i \in A_S[X]$.*

Proof. At a node we are given a monic $g \in LA_S[X]$; at the root $g = F$. Divide each y_i by g, with remainders of degree $< \deg g$, and inspect their coefficients from the top. Each coefficient is a fraction a/s, and we branch on a: in the child $(I, a; U)$ it lies in J, in the child $(I; U, a)$ it is invertible. If all coefficients lie in J we have a leaf with $h = g$. Otherwise some remainder has its highest coefficient not in J invertible, at degree $r < \deg g$, and Lemma 3 gives a monic member of degree r, with which we start again. The degree decreases, so this terminates. $\square$

By Lemma 4, applied over $R = A_S$, we get at each leaf a monic h with $(h) = LA_S[X]$.

5. A local-global principle

In this section $S_1, \dots, S_m$ are comaximal monoids of A: any choice of $s_i \in S_i$ generates the unit ideal. Let B be an A-algebra and write $B_i = S_i^{-1}B$ and $B_{ij} = (S_iS_j)^{-1}B$, where S_iS_j is the monoid of products. We use two facts about B_i: finitely many elements of B_i can be written $p_1/d, \dots, p_r/d$ with one $d \in S_i$, and $p/1 = 0$ in B_i means $sp = 0$ for some $s \in S_i$.

Lemma 7. *Let $z, w_1, \dots, w_r \in B$. If $z \in (w_1, \dots, w_r)B_i$ for every i, then $z \in (w_1, \dots, w_r)$.*

Proof. Fix i. Clearing denominators in $z = \sum_j a_j w_j$ over B_i gives $d \in S_i$ and $p_j \in B$ with $dz - \sum_j p_j w_j = 0$ in B_i; this difference is killed by some $t \in S_i$. Hence $s_i z = \sum_j a_{ij} w_j$ in B, with $s_i = td \in S_i$ and $a_{ij} = tp_j$. Write $\sum_i \lambda_i s_i = 1$; then $z = \sum_j (\sum_i \lambda_i a_{ij}) w_j$. □

Lemma 8. *Let $P_i \in B_i$ be such that P_i and P_j have the same image in B_{ij} for all i, j. Then there is $g \in B$ with $g = P_i$ in B_i for every i.*

Proof. Write $P_i = p_i/d_i$ with $p_i \in B$ and $d_i \in S_i$. In B_{ij} we have $d_jp_i - d_ip_j = d_id_j(P_i - P_j) = 0$, so there are $u_{ij} \in S_i$ and $v_{ij} \in S_j$ with

$$u_{ij}v_{ij}\,(d_jp_i - d_ip_j) = 0 \quad \text{in } B.$$

Choose such a pair for every ordered pair (i, j) and put

$$r_i = \textstyle\prod_k u_{ik}v_{ki} \in S_i, \qquad s_i = d_ir_i \in S_i, \qquad H_i = r_ip_i.$$

Since u_{ij} divides r_i and v_{ij} divides r_j, the product r_ir_j is a multiple of $u_{ij}v_{ij}$, so $r_ir_j(d_jp_i - d_ip_j) = 0$, that is $s_iH_j = s_jH_i$ in B. Write $\sum_i \lambda_i s_i = 1$ and put $g = \sum_j \lambda_j H_j$. Then

$$s_ig = \textstyle\sum_j \lambda_j s_j H_i = H_i = r_ip_i,$$

and since $s_i = d_ir_i$ is invertible in B_i, $g = p_i/d_i = P_i$ in B_i. □

Both lemmas say that $B \to \prod_i B_i$ behaves like the restriction to a cover. The point of Lemma 8 is that it uses one element of each monoid for each pair, and no common exponent.

6. Proof of Theorem 1

Take the tree of Lemma 6, number its leaves $1, \dots, m$ and let $S_1, \dots, S_m$ be their monoids; they are comaximal by Lemma 2. Let $B = A[X]$, so $B_i = A_{S_i}[X]$. At leaf i we have a monic generator h_i of LB_i.

The local generators agree. Over $B_{ij} = A_{S_iS_j}[X]$, both h_i and h_j are monic generators of LB_{ij}, so they are equal by Lemma 5; if their degrees differ, then $B_{ij} = 0$ and they are equal anyway.

Gluing. By Lemma 8 there is $g \in A[X]$ with $g = h_i$ in every B_i. Then $g \in LB_i$ and $y_j \in (g)B_i$ for all i, j, so Lemma 7, applied twice, gives $g \in L$ and $y_j \in (g)$. Hence $L = (g)$.

Descent. Write $g = \sum_j c_j y_j$ and $y_j = ga_j$, and put $b_i = \sum_j e_{ij}c_j$, so that $x_ig = Fb_i$. Then $Fb_ia_j = x_iy_j = Fe_{ij}$ and $F\sum_i a_ib_i = \sum_i x_iy_i = F$; since F is regular, $b_ia_j = e_{ij}$ and $\sum_i a_ib_i = 1$. This proves Theorem 1.

7. Remarks

The usual way to finish (see [1, Corollary 7.7]) is to choose one element s_i in each monoid such that the generators descend to $A_{s_i}[X]$, and then to patch over the principal localisations, raising all elements to a common power. The organisation above was suggested by the formalisation of the proof in a small dependent type theory. There A_S is a quotient of pairs, and an equation in A_S comes with one element of S as witness. So the data at hand are finitely

many elements of each monoid, and it is simplest to multiply them inside the monoid. Stated for any A-algebra, Lemmas 7 and 8 reduce the proof of the theorem to a short sequence of steps, and are reusable. The formal statements only use the two facts about B_i listed above, so they apply to any family of maps $B \to D_i$ with these properties.

The formalisation also showed that only the equations of Theorem 1 are used, not that e is a projector, and that Lemma 5 must allow different degrees, since equality of the degrees of the local generators cannot be decided. The complete proof is checked, from the definition of polynomial rings and localisations, in about 5 000 lines, in under 3 seconds.

References

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